

A Self-Push-Out Counterexample to Question 1 of arXiv:2302.04514

Candidate counterexample packet

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Status. Candidate full negative answer, likely valid, pending human review. The example satisfies every hypothesis of Question 1 and makes the push-out extension an isometric identity operator.

Source question

González and Martínón ask the following on source PDF page 8, Question 1 [1]:

Suppose that E is an order continuous Banach lattice with a weak unit. Is it true that $\overline{T} \in \text{DN-S}(L_1(\mu), \text{PO})$ implies $T \in \text{DN-S}(E, Y)$?

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[9] Theorem 15]. Moreover, if T is supertauberian then T^{co} is an (into) isomorphism [11] Proposition 6.5.3], and the last fact implies T^{**} tauberian because $(T^{co})^{**} \equiv (T^{**})^{co}$ is injective in this case. $\square$

Question 1. *Suppose that E is an order continuous Banach lattice with a weak unit. Is it true that $\overline{T} \in \text{DN-S}(L_1(\mu), \text{PO})$ implies $T \in \text{DN-S}(E, Y)$?*

Source PDF page 8, including the complete statement of Question 1.

For the inclusion $j : E \rightarrow L_1(\mu)$ and an operator $T : E \rightarrow Y$, the source defines

$$\Delta = \{(Tx, -jx) : x \in E\} \subset Y \oplus_1 L_1(\mu), \quad \text{PO} = (Y \oplus_1 L_1(\mu)) / \overline{\Delta},$$

and

$$\overline{T}f = (0, f) + \overline{\Delta}.$$

An operator from a Banach lattice is disjointly non-singular if its restriction to no closed subspace generated by a disjoint sequence is strictly singular.

Idea of the counterexample

Use the inclusion itself as the operator: $T = j : L_p \rightarrow L_1$. On shrinking disjoint supports, this inclusion loses norm and is not disjointly non-singular. But in the push-out, the two copies of the same dense inclusion produce the full anti-diagonal of $L_1 \oplus_1 L_1$. Quotienting by that anti-diagonal retains only the sum of the two coordinates. The extension $f \mapsto [(0, f)]$ therefore becomes the identity on L_1 .

Counterexample and proof

Theorem 1. *Question 1 has a negative answer. For every $1 < p < \infty$ there are an order continuous Banach lattice E with a weak unit, a Banach space Y , and $T : E \rightarrow Y$ such that $\overline{T} \in \text{DN-S}$ but $T \notin \text{DN-S}$.*

Proof. Fix $1 < p < \infty$, and let

$$E = L_p[0, 1], \quad Y = L_1[0, 1], \quad T = j : E \longrightarrow L_1[0, 1]$$

be the canonical inclusion. The space E is order continuous, the constant function $\mathbf{1}$ is a weak unit, and $j(E)$ is dense in $L_1[0, 1]$.

First we show that $T \notin \text{DN-S}$. Put

$$A_n = (2^{-n}, 2^{-n+1}], \quad x_n = 2^{n/p} \mathbf{1}_{A_n}.$$

Then the x_n are disjoint and $\|x_n\|_p = 1$, whereas

$$\|Tx_n\|_1 = 2^{-n(1-1/p)}.$$

In fact, T is compact on $[x_n]$. If $q = p/(p-1)$ and $w_n = 2^{-n/q}$, then, for finitely supported scalars (a_n) ,

$$\left\| T \sum_{n>N} a_n x_n \right\|_1 = \sum_{n>N} |a_n| w_n \leq \|(a_n)\|_{\ell_p} \left(\sum_{n>N} w_n^q \right)^{1/q}.$$

The last factor tends to zero because $w_n^q = 2^{-n}$. Hence the restriction $T|_{[x_n]}$ is the operator-norm limit of finite-rank operators. It is compact, therefore strictly singular, and the definition of DN-S gives $T \notin \text{DN-S}(E, Y)$.

Now compute the push-out. Since $T = j$,

$$\Delta = \{(jf, -jf) : f \in L_p[0, 1]\}.$$

Density of $j(L_p)$ in L_1 gives

$$\overline{\Delta} = \{(g, -g) : g \in L_1[0, 1]\}.$$

Define

$$\Phi : \text{PO} \longrightarrow L_1[0, 1], \quad \Phi([(a, b)]) = a + b.$$

This is well-defined and bijective because the kernel of $(a, b) \mapsto a + b$ is precisely $\overline{\Delta}$. It is isometric: for every $a, b \in L_1$,

$$\begin{aligned} \|[(a, b)]\|_{\text{PO}} &= \inf_{g \in L_1} (\|a - g\|_1 + \|b + g\|_1) \\ &\geq \|a + b\|_1, \end{aligned}$$

while the choice $g = a$ gives the reverse inequality and hence equality.

Finally,

$$\Phi(\overline{T}f) = \Phi([(0, f)]) = f.$$

Thus $\Phi\overline{T} = I_{L_1}$. The identity on L_1 is disjointly non-singular (indeed, it is an isomorphism on every subspace), so $\overline{T} \in \text{DN-S}(L_1, \text{PO})$.

The other arrow in the push-out is also nondegenerate: $\Phi(\overline{j}y) = \Phi([(y, 0)]) = y$. Therefore $\overline{T}$ really is an extension of T in the paper's sense. We have $\overline{T} \in \text{DN-S}$ but $T \notin \text{DN-S}$, as required. $\square$

Verification report

- All source hypotheses hold: $L_p[0, 1]$ is order continuous for $1 < p < \infty$, and $\mathbf{1}$ is a weak unit.
- The failure of DN-S is verified from the definition, not merely from the decay of one sequence: the restriction to $[x_n]$ is explicitly compact.
- The closure of Δ is exactly the full anti-diagonal because $L_p[0, 1]$ is dense in $L_1[0, 1]$.
- Both inequalities for the quotient norm are explicit, proving an isometric rather than merely isomorphic identification of PO with L_1 .
- Under that identification, both $\overline{T}$ and $\overline{j}$ are the identity. There is no kernel or injectivity loophole.
- No computation, external lemma, or unproved dependency is needed beyond the standard fact that compact operators are strictly singular.

Novelty check and limitations

A bounded search on 21 July 2026 covered the run registry, solution, attempt, and proof-gap indexes; the local source corpus; exact searches for the wording of Question 1; the arXiv id, title, and authors; and web searches combining “push-out” with “disjointly non-singular”. The 2024 journal version retains Question 1, but no later answer was found. This supports, but does not prove, novelty. The recommended human check is the push-out sign convention and the use of the closure of Δ ; both are transcribed directly from the source.

References

- [1] M. González and A. Martínón, *Disjointly non-singular operators: Extensions and local variations*, J. Math. Anal. Appl. **530** (2024), 127685; arXiv:2302.04514, DOI: 10.1016/j.jmaa.2023.127685.